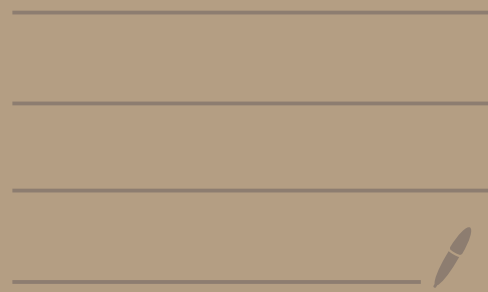
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$$z^l = w^l a^{l-1} + b^l$$

$$a^l = f(z^l)$$

$$\delta_j^l = \frac{\partial C}{\partial z_j^l} = \sum_k \frac{\partial C}{\partial z_k^{l+1}} \cdot \frac{\partial z_k^{l+1}}{\partial a_j^l} \frac{\partial a_j^l}{\partial z_j^l}$$

$$= \sum_k \delta_k^{l+1} w_{kj}^{l+1} f'(z_j^l)$$

$$\frac{\partial C}{\partial w_{ij}^l} = \frac{\partial C}{\partial z_j^l} \frac{\partial z^l}{\partial w_{ij}^l} = \delta_j^l a_i^{l-1} \qquad \frac{\partial C}{\partial b_j^l} = \delta_j^l$$

$$g' + \alpha g = 0 \qquad g(0) = g_0$$

$$g_t = g_0 + x \cdot N(x)$$

$$g_t' = N(x) + xN'(x)$$

$$r(x) = g_t'(x) + \alpha g_t(x)$$

$$= N(x) + xN'(x) + \alpha(g_0 + xN(x))$$

$$C = \frac{1}{N} \sum_n r(x_n)$$

$$N(x) = \sum_j v_j f(w_j x + b_j) + C$$

$$N'(x) = \sum_j v_j w_j f'(w_j x + b_j)$$

$$N''(x) = \sum_j v_j w_j^2 f''(w_j x + b_j)$$